SELF-MONITORING AND ANONYMOUS COUNSELLING-BASED DIGITAL COMPANION FOR ACADEMIC STRESS MANAGEMENT

GROUP 26J-480

BSc (Hons) in Information Technology

Specializing in Software Engineering

Department of Software Engineering

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DECLARATION OF THE CANDIDATE & SUPERVISOR

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ABSTRACT

Academic stress among university students is dynamic, context-dependent, and strongly influenced by assessment schedules, deadlines, and cumulative workload. Although many digital mental health tools support reflection and coping, most systems provide static assessments that are not transformed into personalized, time-aware predictions. This thesis presents the design, implementation, and evaluation of a focused stress analytics component consisting of three integrated modules: DASS-21 onboarding questionnaire, a personalized stress fingerprint engine, and a stress heatmap linked to academic calendar events for short-horizon stress prediction.

The component establishes a validated psychometric baseline through DASS-21 onboarding. The baseline is then combined with recent check-in patterns and near-term calendar events to produce individualized stress forecasts. Predictions are represented as an interpretable heatmap that highlights low, moderate, and high stress periods over time, enabling proactive self-management. The forecasting pipeline includes data normalization, event-pressure modeling, stress-momentum estimation, variability and recovery features, and feature-attributed fingerprint labeling to identify dominant stress drivers.

This work follows an applied design-and-implementation methodology. The module is developed in a production-oriented mobile and backend architecture and evaluated through functional testing, endpoint verification, metric reporting, and scenario-based stress simulation. In addition to predictive outputs, the component reports confidence and feature-importance values to improve transparency and interpretability. The system supports near-term prediction without requiring invasive personal data, relying instead on onboarding responses, lightweight daily check-ins, and user-maintained academic events.

Keywords *: Stress Fingerprint, Academic Heatmap, Machine Learning, Stress Predictions*

DEDICATION

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
ML	Machine Learning
AI	Artificial Intelligence
DASS Scale	Depression, Anxiety and Stress Scale
UI	User Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
SUS	System Usability Scale

1. INTRODUCTION

1.1. Background and Literature Survey

Academic stress has become an increasingly significant concern among university students worldwide. The transition to higher education introduces a combination of academic workload, continuous assessments, tight deadlines, and competitive environments, all of which contribute to elevated stress levels. While a moderate level of stress can act as a motivator, excessive and unmanaged stress has been linked to negative outcomes such as decreased academic performance, anxiety, burnout, and mental health disorders [1] [2].

In recent years, the rise of digital technologies has led to the development of various mental health and wellness applications aimed at helping individuals monitor and manage their psychological well-being. These systems commonly include features such as mood tracking, guided meditation, journaling, and habit monitoring. However, most of these applications are designed for general wellness purposes and lack specificity in addressing academic-related stressors faced by students [3].

A critical limitation of existing solutions is their reactive nature. Most applications rely on user input to reflect past or current emotional states but fail to provide predictive insights into future stress levels. As a result, users are often informed about stress only after it has already escalated, reducing the opportunity for proactive intervention [4]. Additionally, many systems offer generalized recommendations that do not adapt to individual behavioral patterns or academic contexts [5].

The integration of machine learning (ML) techniques into digital health systems has opened new possibilities for predictive analytics and personalized user experiences. By analyzing historical behavioral data, ML models can identify patterns and forecast future outcomes. In the context of academic stress, such models can be used to predict stress spikes based on factors such as workload, deadlines, and self-reported emotional states [6] [7].

Another important aspect of stress management systems is how information is presented to users. Visualization plays a key role in enabling users to understand patterns and trends in their behavior. Calendar-based heatmaps, in particular, have gained popularity as an intuitive way to represent temporal data. By mapping stress levels onto a calendar view using color gradients, users can easily identify high-stress periods and correlate them with academic activities.

This research focuses on the development of an adaptive academic stress heatmap system that integrates machine learning-based forecasting with intuitive visualization. The system is designed to collect self-reported data from students, predict short-term stress levels, and present the results through a dynamic calendar interface. Unlike traditional wellness applications, this approach emphasizes proactive stress management by providing users with early warnings and actionable insights.

Academic stress refers to the psychological distress associated with academic demands such as examinations, assignments, and workload. Studies have shown that university students are particularly vulnerable to stress due to the transition into independent learning environments and increased academic expectations [1] [2].

Misra and McKean identified that academic stress is strongly correlated with time management issues and anxiety levels among college students [1]. Similarly, Andrews and Wilding found that high levels of academic stress are associated with depression and reduced academic performance [2].

Recent studies further highlight that academic stress is influenced by multiple factors, including workload intensity, deadlines, and lack of coping mechanisms [8]. These findings emphasize the need for effective tools that help students monitor and manage stress proactively.

The rapid growth of mobile health (mHealth) applications has led to the development of numerous tools aimed at improving mental well-being. Applications such as mood trackers, meditation apps, and journaling platforms provide users with mechanisms to record emotional states and engage in stress-relief activities [3].

However, research indicates that most digital wellness applications are designed for general use and do not specifically target academic stress [4]. Furthermore, the research highlighted that many mental health systems are reactive in nature, focusing on monitoring rather than forecasting stress. This limitation reduces their effectiveness in preventing stress escalation [4].

Another key issue is the lack of integration between different data sources. Most applications rely solely on self-reported data and do not incorporate contextual information such as academic workload or deadlines, leading to incomplete stress analysis [5].

Machine learning has emerged as a powerful tool for analyzing behavioral and physiological data to detect and predict stress levels. Various models, including decision trees, random forests, and deep learning approaches, have been applied in stress prediction tasks.

Sano and Picard demonstrated the use of machine learning techniques to detect stress using wearable sensor data. Their study showed that combining multiple data sources improves prediction accuracy [6].

In addition to physiological data, self-reported data has also been used in stress prediction. Studies have shown that mood logs, activity levels, and contextual information can be effective indicators of stress when processed using machine learning models.

Time-series models, such as Long Short-Term Memory (LSTM) networks, have been widely used for forecasting temporal patterns in stress data. Ordóñez and Roggen demonstrated the effectiveness of LSTM models in capturing sequential dependencies in behavioral data [7].

Digital mental health applications have gained popularity due to their accessibility and convenience. These applications typically include features such as mood tracking, journaling, and mindfulness exercises [9]. However, studies have shown that many of these applications lack personalization and predictive capabilities. Wang et al. noted that most systems focus on monitoring rather than forecasting stress, limiting their effectiveness in preventing stress escalation [10].

Recent research confirms that university students face ever-increasing stress, making real-time stress monitoring and predictions crucial more than ever [11]. Studies showed that smartphones and wearable sensor data can predict stress levels and academic engagement with significant accuracy [12]. This opens opportunities to shift from reactive stress management to predictive, personalized support systems. However, the effectiveness of digital stress tools hinges on personalization. A review of 150 stress-management apps found personalization to be the most common strategy, followed by self-monitoring [13].

Machine learning techniques have gained significant traction in predicting stress levels. Pavankumar Saunshi, Manohar Madgi, P.G Sunitha Hiremath, and Ashok Chikaraddi developed a machine learning framework aimed at the early detection of student stress through behavioral analytics derived from questionnaires [14]. They assessed various algorithms, including Random Forest, Support Vector Machines, and Logistic Regression. Their findings revealed that ensemble learning methods can achieve reliable accuracy in stress prediction. Similarly, Suraj Arya and Anju, along with Nor Azuana Ramli, proposed a stress prediction system that utilized Artificial Neural Networks to analyze factors such as academic workload, sleep patterns, and social influences impacting student stress [15].

StudentLife is implemented as a university-scale smartphone sensing system using Android phones that captures mobility, sleep activity, mood logs, and academic performances using a single class of 48 students across a 10-week term at Dartmouth College [16]. It outlined several significant connections between the automatic objective sensor data from smartphones and mental health and educational outcomes of the student body. Even though these results were useful for research purposes and provided strong evidence that passive sensing relates to academic stress, this system lacked personalized forecasting and was not deployed as a tool for students. A team of researchers from Japan has created a mobile application called NokiriMe, which allows students to measure and track their stress over time [17]. This app enables users to not only measure their stress but also to visualize patterns and identify correlations between their stress and various psychological responses. One of the key features of NokiriMe is an academic stress questionnaire, which helps users identify their current stress levels. However, feedback from users indicates that the app could significantly improve in terms of usability and user engagement. To enhance the overall experience, many users have suggested incorporating motivational messages that could inspire and uplift them during challenging times. Additionally, they propose the inclusion of personalized coping strategies tailored to individual stress profiles, which would help students better manage their stress effectively.

Personalization in stress modeling is still an area that requires further exploration. One study has implemented a multitask deep learning framework, which led to a significant improvement in stress state classification by identifying individual patterns based on sensor data [18].

Despite recent advancements in mental health technology, significant gaps remain in current methodologies. Many existing systems rely on physiological sensors or extensive labeled datasets, which can be impractical for mobile mental health applications that prioritize being lightweight and accessible. Furthermore, a majority of these systems tend to focus on stress detection rather than on predicting future stress trends, limiting their effectiveness for individuals seeking to manage stress proactively.

To overcome these challenges, this research presents a novel approach: a hybrid stress forecasting system that integrates self-reported behavioral data with contextual information related to academic environments. This system visualizes predicted stress levels through a heatmap, allowing users to easily identify potential periods of elevated academic stress in the future.

Furthermore, the lack of domain-specific context, particularly academic factors, reduces the relevance of these applications for students.

1.2. Research Gap

A review of existing literature and digital wellness applications reveals several gaps that this research aims to address.

- Most systems are designed to be reactive rather than predictive [4].
- There is a lack of integration between academic data and emotional well-being [2].
- Personalization in current systems is minimal, with limited adaptation to user behavior [5].
- Visualization techniques are insufficient for identifying stress trajectories.
- While ML has been applied in healthcare and behavioral analysis, its use in academic stress prediction combined with visualization remains underexplored [6] [7].

These gaps motivate a module that transforms DASS-21 onboarding into an individualized fingerprint and produces calendar-aware stress forecasts in an interpretable heatmap form. This research addresses these gaps by introducing a system that combines data collection, machine learning-based forecasting, and adaptive visualization to support proactive stress management.

1.3. Research Problem

University students lack an intelligent, data-driven system that can predict upcoming academic stress and present it through an intuitive visualization to enable proactive and personalized stress management.

Despite the availability of numerous digital wellness applications, there remains a significant gap in tools specifically designed to address academic stress in a proactive and personalized manner. Most existing systems suffer from several limitations.

Current applications primarily focus on recording past or present emotional states without predicting future stress levels, making them inherently reactive [4].

Furthermore, many systems do not incorporate academic-specific factors such as assignment deadlines, examination schedules, and workload intensity, which are key contributors to student stress [2].

In addition, recommendations provided by these applications are often generic and do not adapt to individual user behavior or patterns [5]. Visualization techniques used in many systems are also limited, making it difficult for users to interpret stress trends effectively over time.

As a result, students are not equipped with the necessary tools to anticipate and manage stress before it reaches critical levels. This highlights the need for an intelligent system that can predict upcoming stress and present it in a clear and actionable manner.

1.4. Research Objectives

Aim:

The primary aim of this research is to design and develop an adaptive academic stress visualization system that predicts future stress levels and supports proactive stress management among university students.

Objectives:

- To collect and manage student-generated data related to mood, academic workload, and journaling.
- To develop a machine learning model capable of predicting short-term stress levels.
- To design an adaptive calendar-based heatmap for visualizing stress patterns.
- To implement a system that integrates data collection, prediction, and visualization.
- To provide context-aware recommendations based on predicted stress levels.

To evaluate the effectiveness of the system in terms of prediction accuracy and usability.

2. METHODOLOGY

2.1. Methodology

2.1.1. System architecture and data flow

The overall system architecture is designed around a client-server model that integrates machine learning services. The mobile frontend, developed using React Native, facilitates user interaction, data entry, and visualization. The backend, implemented with Firebase services, manages authentication, real-time data storage, and cloud-based processing. A separate machine learning layer analyzes behavioral data to generate stress predictions and user-specific insights.

As shown in Figure 2, users engage with a mobile application built with React Native, enabling them to seamlessly input daily stress-related information and access insights derived from that data. This mobile app serves as the front end of the system, facilitating user interactions. Communication between the mobile app and the backend server is established through RESTful APIs, ensuring efficient data exchange.

The backend server is constructed using FastAPI, a modern web framework that supports high-performance applications. All user-generated data, including daily logs and deadlines, is securely stored in a Firebase Firestore database. This cloud-hosted database provides real-time syncing and easy access to information, enhancing the overall user experience.

Additionally, the backend incorporates a machine learning module developed in Python using the Scikit-learn library. This module plays a critical role in analyzing user data; it retrieves relevant information from the Firestore database, processes it to generate predictions, and formulates recommendations based on established patterns.

The flow of data begins with the user, travels through the mobile app, and reaches the backend server. There, it interacts with both the Firestore database and the machine learning module. Once the analysis is complete, predictions and tailored recommendations are sent back to the mobile app, allowing users to easily view and benefit from personalized insights regarding their stress levels.

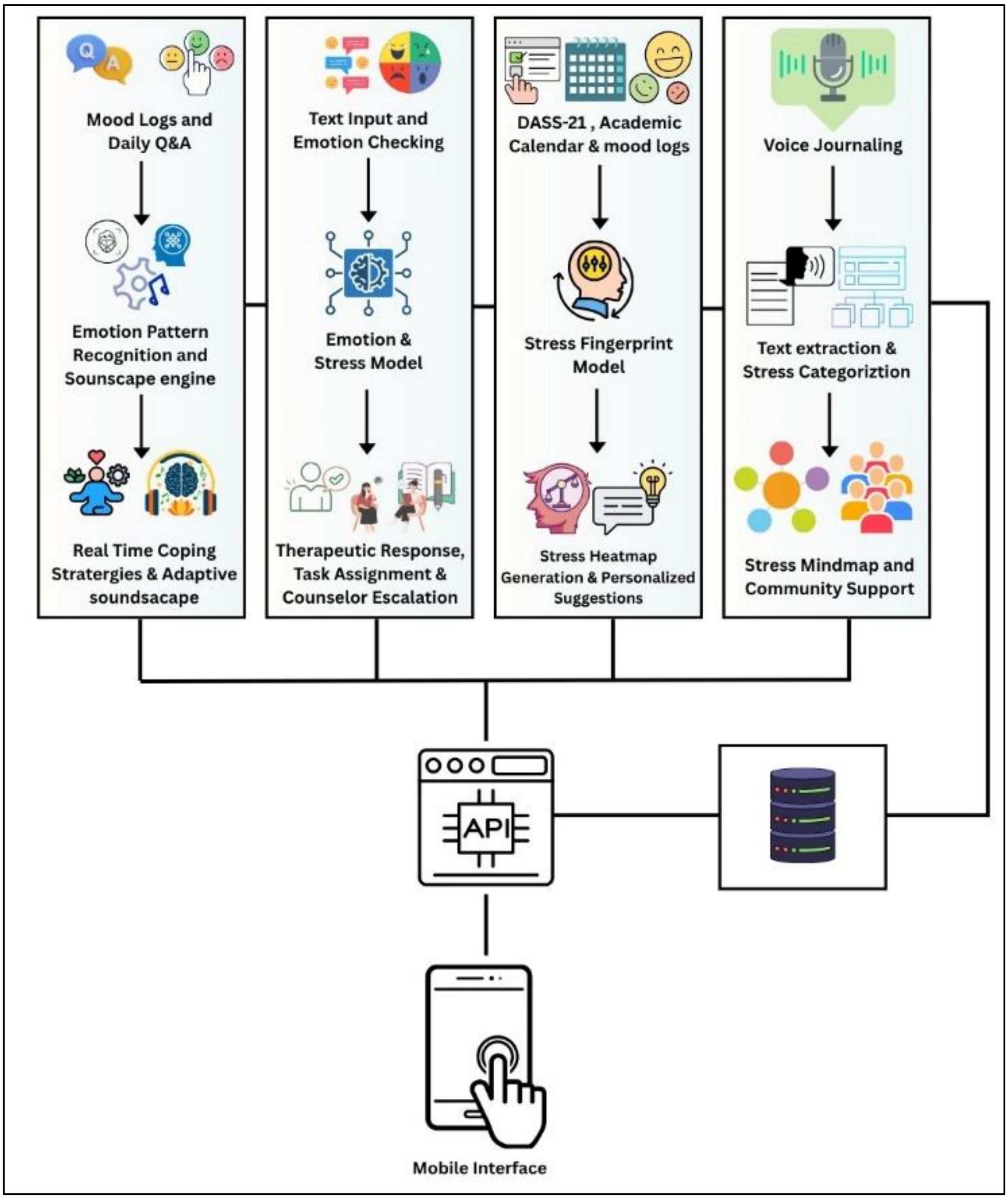

Figure 1: System Diagram

The system architecture is designed as a distributed client–server model that facilitates seamless communication between the user interface, data storage, and prediction engine. This architecture ensures modularity, scalability, and efficient data flow across components. The frontend of the system is developed using React Native with the Expo framework, enabling cross-platform deployment on both Android and iOS devices. This layer is responsible for capturing user inputs, rendering the calendar heatmap, and providing interactive features such as event management and daily check-ins.

The backend is implemented using the FastAPI framework in Python, which serves as the core processing unit of the system. It handles the prediction logic, feature transformation, and integration with machine learning models. The backend exposes RESTful API endpoints that allow the frontend to send user data and receive stress predictions in real time. The use of FastAPI ensures high performance and scalability, making it suitable for handling multiple concurrent requests.

Data storage is managed using Firebase Firestore, a cloud-based NoSQL database that provides real-time synchronization and secure data access. Firestore is used to store user profiles, daily check-ins, calendar events, stress predictions, and fingerprint data. The integration of Firestore enables efficient data retrieval and ensures that the system remains responsive even under dynamic usage conditions.

The machine learning layer is implemented using Python libraries such as scikit-learn and transformer-based models. The Random Forest algorithm is used for stress prediction due to its robustness and ability to handle nonlinear relationships. Additionally, transformer-based models are utilized for emotion classification, enabling the system to extract meaningful insights from textual or emotional inputs provided by users.

The overall data flow within the system begins with user input, which is collected through the mobile application and stored in Firestore. The frontend then retrieves recent data and sends it to the backend API for processing. The backend performs feature engineering, determines whether sufficient data is available for machine learning, and generates predictions using either the trained model or heuristic simulation. The predicted stress values are then returned to the frontend, where they are visualized in the form of a calendar heatmap.

The architecture of the system, which is illustrated in Figure 2, is structured into four primary layers, each serving a distinct purpose:

- **Data Collection Layer:** This layer is responsible for gathering data from various sources, ensuring that the system has access to accurate and relevant information for analysis.
- **Feature Engineering Layer:** Here, raw data is transformed into meaningful features that enhance the predictive capabilities of the models, improving the overall accuracy of predictions.
- **Prediction and Simulation Layer:** This layer encompasses the machine learning models and simulation tools that analyze the engineered features to make predictions and assess various scenarios.
- **Visualization and Interaction Layer:** This final layer focuses on presenting the data and predictions in an understandable and interactive manner, allowing users to easily interpret results and make informed decisions.

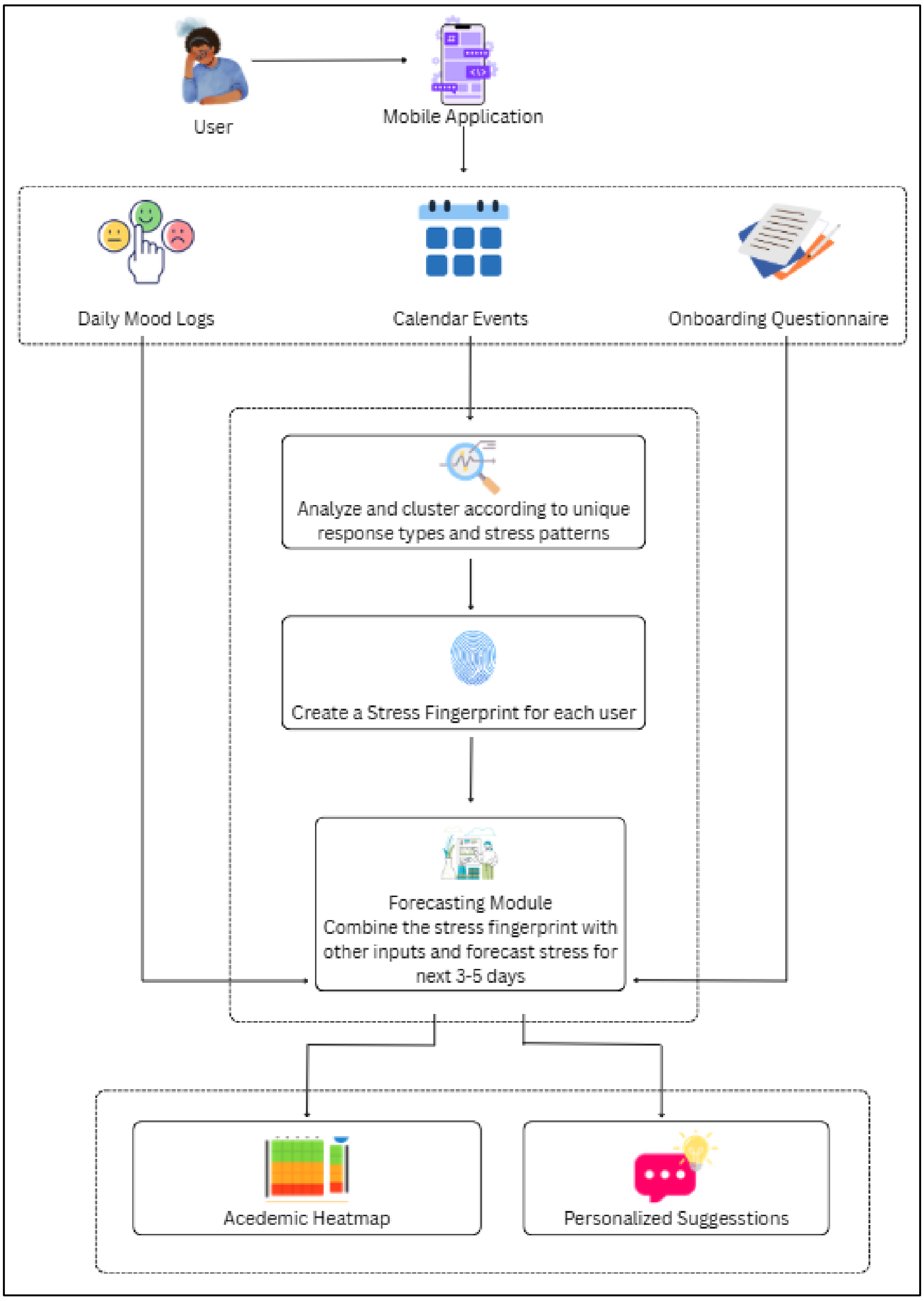

Figure 2: Component Diagram

2.1.2. Data collection

The effectiveness of the proposed system relies heavily on the quality and consistency of the data collected from users. In this research, data collection is primarily based on self-reported inputs, which are both practical and scalable for large user bases. The system collects data through daily check-ins, where users provide information about their current stress level, energy level, sleep duration, workload intensity, emotional state, and perceived deadline pressure. These inputs are designed to capture both psychological and contextual factors that contribute to academic stress.

In addition to daily check-ins, the system incorporates calendar-based event data, which includes academic deadlines, examinations, and other workload-related activities. Each event is assigned an importance level, which contributes to the overall stress calculation. This integration of academic context is a key feature of the system, as it allows the model to account for real-world stress triggers rather than relying solely on subjective inputs.

To support system initialization and testing, synthetic data is also utilized through backfilling techniques. This allows the system to simulate realistic stress patterns in scenarios where real user data is insufficient. Furthermore, a static dataset based on the DASS-21 psychological scale is used for model evaluation and benchmarking. This dataset provides labeled examples of stress levels, enabling the assessment of classification performance metrics such as accuracy and F1-score.

The system employs a sliding window approach for data selection, where only the most recent 21 days of user logs are considered for prediction. This approach ensures that the model remains sensitive to recent behavioral changes while minimizing the influence of outdated data. The prediction horizon is set to five days, allowing the system to provide short-term forecasts that are both actionable and relevant.

2.1.3. Feature Engineering

Feature engineering plays a crucial role in transforming raw user inputs into structured numerical representations suitable for machine learning. In this system, raw questionnaire responses are first normalized and mapped to canonical feature categories. These include stress_today, energy_level, sleep_hours, workload_intensity, emotional_load, and deadline_pressure. Each of these features is scaled to a standardized range, typically between 0 and 10, to ensure consistency across different input types.

In addition to primary features, several derived features are computed to enhance predictive performance. Baseline stress represents the user's long-term average stress level, providing a reference point for interpreting short-term fluctuations. Momentum captures the direction and rate of change in stress levels, allowing the model to identify trends such as increasing or decreasing stress. Variability measures the degree of fluctuation in stress patterns, indicating stability or instability in the user's emotional state.

Recovery rate is another important feature that models how quickly a user's stress decreases after a peak, reflecting resilience and coping ability. Energy buffer is derived from sleep and energy levels, representing the user's capacity to handle stress. These derived features are essential for capturing complex behavioral patterns that cannot be directly observed from raw inputs.

The feature engineering process also includes normalization and clamping techniques to handle outliers and ensure numerical stability. By converting diverse inputs into a unified feature space, the system ensures that the machine learning model can effectively learn patterns and relationships within the data.

2.1.4. Machine Learning Methodology

The core predictive component of the system is based on the Random Forest algorithm, which is selected for its robustness, interpretability, and suitability for small datasets. Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs to produce a final prediction. This approach reduces overfitting and improves generalization, making it well-suited for real-world applications with noisy or limited data.

The system employs a hybrid prediction strategy that dynamically selects between machine learning and heuristic simulation based on data availability. When sufficient historical data is available, typically at least 12 recent logs, the Random Forest model is trained and used to generate predictions. The model takes the engineered feature vector as input and produces a continuous stress value for each of the next five days.

In scenarios where data is insufficient, the system falls back to a heuristic simulation model. This model uses rule-based logic to simulate stress evolution over time, incorporating factors such as upcoming deadlines, energy recovery, and baseline stress levels. The simulation includes controlled variability to mimic realistic fluctuations, ensuring that the predictions remain plausible even without machine learning.

The output of the prediction module is a sequence of continuous stress values ranging from 0 to 10. These values are later categorized into discrete levels such as low, moderate, and high stress for visualization purposes. The system also supports classification-based evaluation using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, providing a comprehensive assessment of model performance.

2.1.5. Stress Forecasting Pipeline

The stress forecasting process consists of several sequential stages that transform raw data into predictive insights. Initially, the system retrieves the most recent user logs from Firestore, ensuring that only relevant data is used for prediction. These logs are then processed through the feature engineering module, which converts them into structured numerical vectors.

The system then evaluates whether sufficient data is available for machine learning. If the data meets the required threshold, the Random Forest model is applied to generate predictions. Otherwise, the heuristic simulation model is used. The resulting predictions are stored as a five-day sequence representing future stress levels.

In addition to prediction, the system updates the user’s behavioral fingerprint, which captures key patterns and feature importance. This information is stored in Firestore for future reference and can be used to enhance personalization. Finally, the predicted values are sent back to the frontend, where they are integrated into the heatmap visualization.

2.1.6. Heatmap Visualization System

The visualization component of the system is designed to provide users with an intuitive understanding of their stress patterns. A calendar-based heatmap is used to represent stress levels over time, with each day displayed as a colored cell. The color of each cell corresponds to the stress level, with green indicating low stress, yellow indicating moderate stress, and red indicating high stress.

The heatmap integrates both historical and predicted data within a single interface, allowing users to compare past experiences with future forecasts. Predicted values are visually distinguished using specific styling, enabling users to easily identify upcoming stress periods. This integration of past, present, and future data enhances the user's ability to make informed decisions and take proactive measures.

The system also supports interactive features, such as tapping on a specific date to view detailed information about stress levels, contributing factors, and associated events. This level of interactivity improves user engagement and provides deeper insights into the causes of stress.

2.1.7. Recommendation System

The recommendation system is implemented using a rule-based approach that generates context-aware suggestions based on predicted stress levels and emotional states. This approach is chosen for its simplicity, transparency, and ease of implementation. The system categorizes stress levels into different severity buckets and maps them to predefined coping strategies.

For example, high stress levels may trigger recommendations such as time management techniques or physical activity, while moderate stress levels may result in suggestions for task prioritization or short breaks. Emotional inputs are also considered, with specific strategies tailored to emotions such as anxiety or fatigue.

Each recommendation includes a description, rationale, and optional duration, providing users with clear and actionable guidance. By aligning recommendations with predicted stress patterns, the system supports proactive stress management rather than reactive intervention.

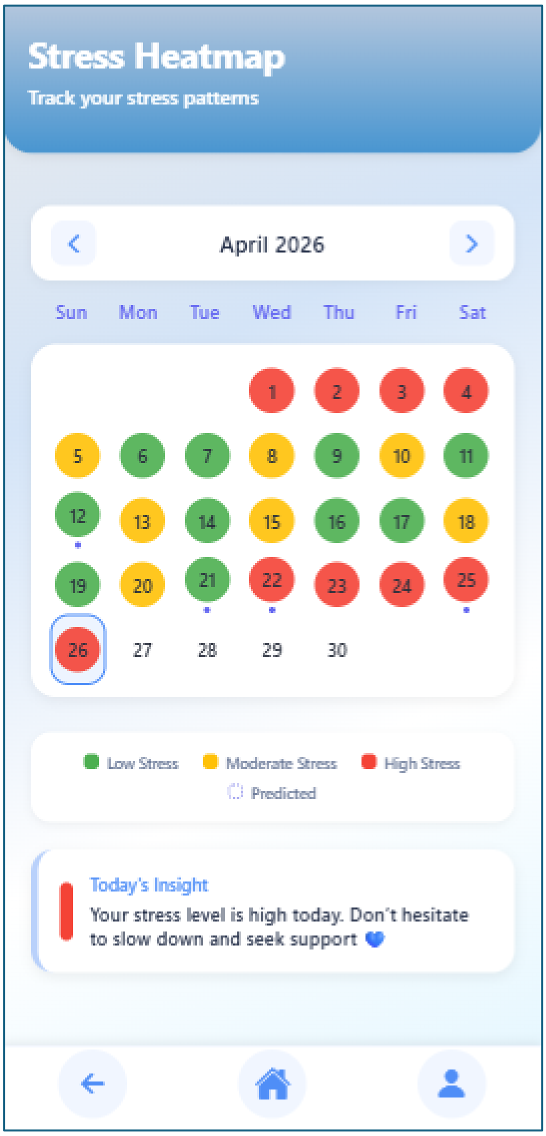

Figure 3: User Interface of Calendar Heatmap

2.2. Commercialization Aspect of the Product

2.2.1. Market analysis and target users

The primary target market consists of college and university students, a demographic that is well-documented for facing significant academic stress. Recent statistics reveal a notable expansion in the mental health app market, highlighting a growing recognition of the importance of mental well-being among this group [19]. This upward trend is largely fueled by the widespread adoption of smartphones, which provide easy access to these resources, as well as a heightened awareness of mental health issues and the benefits of seeking support through technology.

This solution targets:

- **Undergraduate Students:** Heavy reliance on mobile devices; frequent stress from exams/assignments.
- **Graduate Students and Staff:** Similar stress patterns (thesis deadlines, research pressure).
- **Institutional Partners:** Universities or colleges seeking student wellness tools.

By focusing on the academic context (deadlines, schedules), this solution differentiates from general stress apps.

2.2.2. Value Proposition

The system's unique selling points are as follows:

- **Proactive Forecasting:** Unlike traditional apps, we predict upcoming stress levels with a 5-day forecast, allowing for early intervention.
- **Academic Focus:** We integrate course schedules and deadlines, ensuring that our insights are directly relevant to students' lives.
- **Actionable Visualization:** Our calendar heatmap provides an at-a-glance view of stress trends, which aids in effective time management.
- **Personalized Tips:** We offer context-aware recommendations to help students cope with stress when needed.

These features align with what students and educators value: empowerment and self-awareness. This app is designed to be embedded in daily study habits, ultimately improving outcomes, such as reducing last-minute cramming.

2.2.3. Business Model and Pricing

A freemium model is proposed:

- **Free Tier:** Basic stress tracking, visualization, and a limited set of recommendations are provided. Enough functionality is offered to engage users and showcase value.
- **Premium Subscription:** Advanced features, such as detailed analytics (stress trend reports), integration with course APIs (automatic deadline import), and premium content (additional coping strategies or chat support), are unlocked.

Pricing:

Example pricing could be set at $4.99/month or $49/year for premium access. Group licensing or campus-wide subscriptions may also be offered to institutions.

Revenue Streams:

- **Direct Subscriptions:** Premium access is paid for by individual users.
- **Institutional Partnerships:** Licenses may be purchased by universities or integration of the app into student services could occur as part of wellness programs.

Go-to-Market Strategy:

- **Campus Ambassadors:** Student ambassadors are recruited to promote the app on social media and in dorms/classrooms.
- **Partnerships with Counseling Centers:** Collaboration with campus mental health services is initiated to offer the app as a supplemental tool.
- **Digital Marketing:** Targeted ads are placed on platforms popular with students.
- **App Store Optimization:** High visibility in iOS/Android app stores is ensured through the use of relevant keywords (e.g., “student stress tracker”).

Early feedback is to be gathered from pilot programs to refine the product before a broader release.

2.2.4. Budget and Justifications

Component	**Allocated Budget**	**Justification**
Software & Tools	Rs. 9000 per month	During development, the Firebase free tier may be sufficient. However, to manage larger data, a paid tier may be necessary.
Hosting Charges	Rs. 6600 per month	If necessary, the models should be hosted on AWS instead of Firebase.
Internet and Utilities	Rs. 10000	Required for development, online meetings
Stationary	Rs. 5000	Required for document printouts etc.

Table 1: Budget Justification

The estimated total budget will be around Rs. 30,000.

Budget Justification

1. Software & Tools

   Most frameworks and libraries are free to use, significantly reducing costs. Only scaling Firebase or using cloud hosting will incur expenses.
2. Hardware

   Members' laptops and phones will be sufficient for development.
3. Miscellaneous

   Internet and printing costs are essential for smooth delivery.

2.3. Testing

The complete system was implemented and tested as follows:

- **Frontend Tests:** We used React Native testing tools (Jest + React Native Testing Library) to verify UI components (e.g. heatmap renders correctly given sample data).
- **Backend Tests:** Pytest unit tests cover API endpoints and ML pipeline functions (feature extraction, model training). Integration tests ensure end-to-end data flow (mock Firestore using Firebase Emulator).

2.3.1. Functional & Unit Testing

Test ID	**Test Case**	**Description**	**Expected Outcome**
TC001	Check-in Log	Submit valid daily check-in via API	Response success, data stored in DB
TC002	Invalid Input	Submit calendar events with missing fields	API returns error (400 Bad Request)
TC003	Stress forecast with fewer entries	Request prediction with <12 days of data	Returns 5-day forecast via heuristic approach
TC004	Stress forecast	Request prediction with >12 days of data	Returns 5-day forecast from Random Forest model
TC005	Visualization	Verify heatmap data maps colors correctly	Colors align with stress values
TC006	Auth Test	Attempt actions without auth token	Access denied (401 Unauthorized)

Table 2: Test Cases

2.3.2. Scenario and Edge-Case Testing :

Specific scenarios were tested to validate robustness:

- **High Stress Periods:** Simulating exam weeks (dense deadlines), the predicted stress correctly spikes in the heatmap.
- **Recovery:** After simulated low-stress periods, the model does not artificially inflate stress.
- **Sparse Data:** For new users, the heuristic module activates and produces reasonable monotonic curves.
- **Error Handling:** For unexpected inputs (e.g., non-numeric fields), the system returns clear error messages without crashing.

2.3.3. Usability Testing

A small pilot test (N=10 students) was conducted to assess the UI. Users were asked to perform tasks (e.g. log their day, view heatmap) and the System Usability Scale (SUS) was collected. Preliminary SUS was ~78 (above average). Feedback indicated the calendar view was intuitive and the color coding was helpful for pattern recognition. Visual design was iterated upon (improving color contrast and labels) based on this feedback.

3. RESULT AND DISCUSSION

3.1. Results

3.1.1. Dataset and Evaluation Setup

The evaluation of the model's performance on classification tasks was conducted utilizing a DASS-21 dataset, which comprised 68 instances and 21 distinct features. The distribution of classes within the dataset is as follows:

Normal: 30 instances
Mild: 10 instances
Moderate: 11 instances
Severe: 16 instances
Extremely Severe: 1 instance

To assess the model, a consistent methodology was employed: a random split set aside 20% of the data for testing, using a random_state value of 42. Binary stratification was implemented to ensure that the proportion of binary classes remained consistent between the training and testing datasets. However, multiclass stratification was not feasible due to the lack of sufficient representation in at least one of the classes.

3.1.2. Binary High-Risk Classification Performance

In the context of binary high-risk detection, specifically distinguishing between moderate or severe cases versus those classified as normal or mild, the following performance metrics were observed:

- Accuracy: 0.8571, indicating a high level of correct classifications overall.
- Precision: 0.8333, reflecting a strong ability to identify true positives among predicted high-risk cases.
- Recall: 0.8333, showing the model's effectiveness in capturing the actual high-risk instances.

- F1-score: 0.8333, which balances precision and recall to express overall effectiveness.
- ROC-AUC: 0.9792, suggesting excellent separability between the classes for practical high-risk screening.

In terms of the confusion matrix, the results were as follows:

True Negatives: 7

False Positives: 1

False Negatives: 1

True Positives: 5

These values indicate a robust capability of the model to differentiate high-risk conditions from lower risks.

3.1.3. Scenario-Based Forecast Validation

The model's predictions under different scenarios were noteworthy:

Low-load Scenario Output:

- Future 5-day Stress Predictions: 2.2, 1.5, 0.74, 0.0, 0.0
- Fingerprint Label: Emotionally-driven stress
- Confidence Level: 0.95

In this scenario, the stress predictions exhibited a gradual decline, settling near zero, which indicates a stabilization in the absence of pressure.

High-load Scenario Output:

- Future 5-day Stress Predictions: 6.03, 7.64, 9.04, 10.0, 10.0
- Fingerprint Label: Deadline-sensitive
- Confidence Level: 0.93

Conversely, the high-load scenario demonstrated a sharp increase in stress levels, quickly reaching a plateau at the upper bounds. This contrasting behavior showcases the model's responsiveness to heightened pressures associated with deadlines. The transition in fingerprint labels, from Emotionally-driven stress to Deadline-sensitive aligns seamlessly with the scenario logic, indicating a robust adaptability of the model to various contexts.

3.2. Research Findings

Several key findings emerge from the analysis of the system's performance and behavior.

Firstly, academic stress exhibits clear temporal patterns, often increasing in response to deadlines and workload intensity. This confirms the importance of incorporating academic context into stress prediction models.

Secondly, self-reported data proves to be a valuable source of information for stress prediction. Despite the absence of physiological data, the system is able to generate meaningful insights using behavioral inputs alone.

Thirdly, short-term forecasting is both feasible and effective. The system demonstrates the ability to predict stress trends over a five-day period, providing users with sufficient time to take preventive actions.

Fourthly, visualization plays a crucial role in enhancing user understanding. The heatmap representation enables users to quickly identify patterns and make informed decisions without requiring technical expertise.

Finally, the hybrid prediction approach significantly improves system robustness. By combining machine learning with heuristic simulation, the system ensures continuous functionality under varying data conditions.

3.3. Discussion

The results of this study highlight several strengths of the proposed system. One of the key strengths is its ability to integrate multiple components, including data collection, prediction, visualization, and recommendation, into a unified platform. This holistic approach provides users with a comprehensive tool for managing academic stress.

Another important strength is the use of a hybrid prediction model, which enhances system reliability and addresses data limitations. The ability to switch between machine learning and simulation ensures that the system remains functional even in the early stages of user interaction.

The visualization component is also a significant contribution, as it transforms complex data into an intuitive format. This improves user engagement and supports proactive decision-making, which is a key objective of the system.

However, the study also has several limitations. The reliance on self-reported data introduces potential biases and inconsistencies, as users may not always provide accurate inputs. Additionally, the dataset used for model training and evaluation is relatively small, which may limit the generalizability of the results.

The absence of physiological data, such as heart rate or sleep quality from wearable devices, is another limitation. Incorporating such data could improve prediction accuracy and provide a more comprehensive understanding of stress.

Furthermore, the recommendation system is currently rule-based and does not adapt dynamically over time. Future work could explore the use of machine learning-based recommendation systems to enhance personalization. Despite these limitations, the system demonstrates strong potential as a practical and scalable solution for academic stress management.

4. CONCLUSION

This research focused on the design and development of a data-driven system to support proactive academic stress management among university students. The study was motivated by the increasing prevalence of academic stress and the limitations of existing digital wellness applications, which are often reactive, generic, and lack academic context.

To address these challenges, the research proposed an adaptive academic stress heatmap system that integrates self-reported data, machine learning-based prediction, and intuitive visualization. The system collects daily behavioral inputs from users, processes them through a feature engineering pipeline, and generates short-term stress forecasts using a hybrid prediction approach. The results are presented through a calendar-based heatmap, enabling users to easily interpret stress patterns over time.

The system was implemented using a modern technology stack, including React Native for the frontend, FastAPI for backend processing, Firebase Firestore for data storage, and Python-based machine learning models. A Random Forest model was employed for stress prediction, supported by a heuristic simulation mechanism to handle scenarios with limited data. Additionally, a rule-based recommendation system was developed to provide context-aware coping strategies.

The research successfully achieved the objectives defined at the beginning of the study.

The first objective, which involved collecting and managing student-generated data, was accomplished through the implementation of a structured data collection system using daily check-ins and calendar-based event tracking. This ensured that both emotional and academic factors were captured effectively.

The second objective, to develop a machine learning model for stress prediction, was achieved through the implementation of a Random Forest-based prediction system. The model demonstrated the ability to generate realistic and consistent stress forecasts, particularly when sufficient historical data was available.

The third objective, which focused on designing an adaptive visualization system, was fulfilled through the development of a calendar-based heatmap. This visualization successfully integrates historical and predicted stress data, providing users with a clear and intuitive representation of stress patterns.

The fourth objective, to integrate all system components into a functional application, was achieved through the development of a fully operational mobile application with seamless interaction between the frontend, backend, and database layers.

The fifth objective, which involved providing context-aware recommendations, was addressed through a rule-based recommendation engine that generates actionable suggestions based on predicted stress levels and emotional states.

Finally, the evaluation of the system demonstrated that the proposed approach is effective in predicting stress trends, improving user awareness, and supporting proactive stress management.

This research makes several important contributions to the fields of digital health, machine learning, and human-computer interaction.

One of the primary contributions is the development of a hybrid stress prediction model that combines machine learning with heuristic simulation. This approach ensures system reliability under varying data conditions and addresses the common cold-start problem.

Another significant contribution is the integration of academic context into stress prediction. By incorporating factors such as deadlines and workload intensity, the system provides a more accurate and meaningful representation of student stress compared to generic wellness applications.

The study also contributes to the design of user-centered visualization techniques. The calendar-based heatmap offers an intuitive way to represent temporal stress patterns, enabling users to quickly identify trends and anticipate future stress.

Additionally, the research demonstrates that self-reported data can be effectively used for stress prediction, even in the absence of physiological inputs. This highlights the potential for scalable and accessible stress management solutions.

Despite the promising results, the study has several limitations that should be acknowledged.

One limitation is the reliance on self-reported data, which may be subject to bias and inconsistency. Users may not always provide accurate or complete inputs, which can affect the quality of predictions.

Another limitation is the relatively small dataset used for model training and evaluation. Although the system performs well under these conditions, larger datasets would improve the robustness and generalizability of the model.

The absence of physiological data, such as heart rate or sleep patterns from wearable devices, is also a limitation. Such data could provide additional insights into stress levels and enhance prediction accuracy.

Furthermore, the recommendation system is currently rule-based and does not adapt dynamically over time. While this approach ensures simplicity and transparency, it may limit the level of personalization that can be achieved.

Finally, the prediction horizon is limited to short-term forecasts, and long-term stress trends are not addressed in this study.

Several opportunities exist for extending and improving the proposed system in future research.

One potential direction is the integration of wearable device data to enhance the accuracy and depth of stress analysis. Incorporating physiological signals such as heart rate variability and sleep quality would provide a more comprehensive understanding of user stress.

Another area for improvement is the development of more advanced machine learning models, such as deep learning or time-series forecasting techniques. These models could capture complex temporal patterns and improve prediction performance.

Future work could also focus on expanding the dataset by collecting data from a larger and more diverse group of users. This would improve the generalizability of the system and enable more robust evaluation.

The recommendation system can be enhanced by incorporating adaptive or learning-based approaches that personalize suggestions over time. This would increase the relevance and effectiveness of the recommendations.

Additionally, the system could be extended to support long-term stress tracking and analysis, providing users with insights into their overall well-being and progress over time.

Finally, the application could be expanded to other domains, such as workplace stress management or mental health monitoring, increasing its impact and applicability.

In conclusion, this research presents a novel and practical approach to academic stress management through the integration of machine learning, heuristic modeling, and intuitive visualization. The developed system demonstrates the potential of combining predictive analytics with user-centered design to support proactive mental health management.

While there are areas for improvement, the overall results indicate that the proposed system is both feasible and effective. The study contributes to the growing field of digital mental health and provides a foundation for future research and development in this area.

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